Retail Business Performance & Profitability Analysis

Elevate Labs Project Phase | Portfolio learning project

Introduction

This project analyzes retail transaction data to understand revenue, profit, category and regional performance, discount patterns and inventory holding days. The dataset is synthetic and does not represent a real retailer.

Abstract

The cleaned dataset contains 1,600 transaction lines, 917 distinct orders and 554 distinct customers. Total sales are 1,519,413.46, total profit is 702,972.17, and profit margin is 46.27%. Values are illustrative USD.

Tools Used

Python (Pandas, NumPy, Matplotlib, Seaborn), MySQL 8, Jupyter Notebook, CSV and GitHub. A browser-based interactive dashboard is included.

Steps Involved

1. Generated and inspected transaction-line records with dates, product, region, quantity, price, cost, discount and inventory days.
2. Removed exact duplicate rows and imputed missing region and discount values using mode and median.
3. Calculated sales, profit, margin, order/customer counts and category, sub-category, regional and monthly summaries.
4. Prepared MySQL queries for KPIs, profitability, discount bands, inventory, monthly trends, a view and indexes.
5. Used Python charts and correlation analysis to explore profitability drivers and prepare dashboard-ready outputs.

Selected KPIs

Metric	Value
Total sales	1,519,413.46
Total profit	702,972.17
Profit margin	46.27%
Orders	917
Customers	554

Conclusion & Recommendations

In this generated sample, Technology has the highest category profit, Labels is the lowest-profit sub-category, and South leads regional sales. A retailer could review low-profit sub-categories, discount approval rules and products with long inventory days. Validate these ideas with actual operational data before making decisions.

Limitation: Synthetic practice data only; no real-world business impact is claimed. Correlation does not establish causation.